

Fundamental Quantum Limits of Power-Constrained Multi-Span Fiber-Optic Transmission: HCF vs Conventional Fibers

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Abstract We explore fundamental quantum limits on the system capacity of multi-span HCF and standard silica-based fiber links in the power-constrained regime, comparing phase-sensitive and phase-insensitive amplifiers and Shannon vs Gordon-Holevo capacity. ©2026 The Author(s)

Introduction

The emerging hollow core fibers (HCF) hold an enormous promise for future optical communications due to their very high nonlinear threshold, low latency, low dispersion, and low loss [1]. HCF samples with 0.04 dB/km loss have been recently reported [2] and multi-span longhaul HCF transmission with 200+km spans has already been demonstrated [3]. In power-constrained multi-span transmission, e.g. submarine links, the total delivered power is divided among many parallel optical paths to optimize the cost/bit, where an optical path may correspond to a single-mode fiber strand, and/or a core and/or a guided mode thereof. Note that HCF is essentially single-mode, whereas various types of multi-mode and/or multi-core silica-based fibers have been reported in the literature [4]. Moreover, the noise figure (NF) of conventional phase-insensitive amplifiers (PIA), e.g. EDFAs, is lower bounded by 3 dB due to quantum mechanical effects, while in phase-sensitive amplifiers (PSA), one of the two quadratures of the optical field can be amplified with an NF approaching 0 dB [5]. Finally, in the regime of very low signal power per optical path, the difference between the conventional Shannon capacity and the general quantum mechanical Gordon-Holevo bound should be investigated [6] and the effect of signal droop should be studied [7]. All these points call for a revised analysis of the performance of future power-constrained multi-span systems based on HCF and their fundamental achievable capacities with PIA and PSA, which we propose to do, for the first time to our knowledge, in this work.

Methodology

As shown in Fig. 1 we consider an optical fiber-based communication link of total length L composed of M parallel paths. Each optical path is composed of identical spans of length ΔL and the fiber type of all spans is either HCF or conventional silica-based. The fiber loss coefficient is denoted

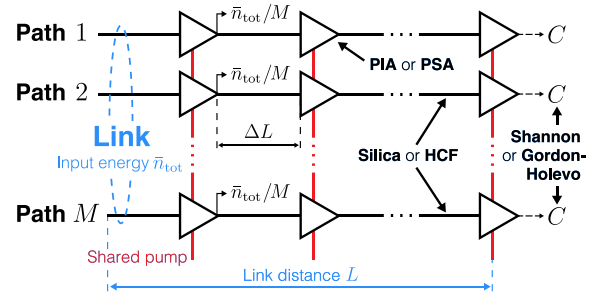


Fig. 1: Schematic link model of total distance L and input energy $\bar{n}_{\text{tot}}$ divided equally over M parallel multi-span fiber paths, either silica-based or hollow core (HCF). After each span of length ΔL either phase-insensitive (PIA) or phase-sensitive (PSA) amplification is applied. Transmission performance is measured by output capacity, either conventional (Shannon) or quantum-optimal (Gordon-Holevo).

by α and the total average number of input photons per WDM channel over all parallel paths is denoted by $\bar{n}_{\text{tot}}$ for one use. In this work we focus on a single WDM channel; the extension to multichannel systems is straightforward once the exact spectral profile of the fiber loss is provided. The span loss is compensated either by a PIA or a PSA. The PIA amplifies both optical quadratures x^Q, x^I by a factor $\sqrt{G}$ while adding excess amplified spontaneous emission (ASE) noise proportional to $G - 1$ [8, 9]. The relation between variances of input and output optical quadratures $S_{\text{in/out}}^{Q/I} = \text{Var}[x_{\text{in/out}}^{Q/I}]$ reads

$$S_{\text{out}}^{Q/I} = G e^{-\alpha \Delta L} S_{\text{in}}^{Q/I} + G - (1 + G e^{-\alpha \Delta L}) / 2. \quad (1)$$

On the other hand, PSA amplifies one quadrature by a factor $\sqrt{G}$ at the cost of attenuating the other one by $1/\sqrt{G}$, while adding noise asymmetrically to both quadratures as follows [10]

$$S_{\text{out}}^Q = G e^{-\alpha \Delta L} S_{\text{in}}^Q + G (1 - e^{-\alpha \Delta L}) / 2, \quad (2)$$

$$S_{\text{out}}^I = e^{-\alpha \Delta L} S_{\text{in}}^I / G + (1 - e^{-\alpha \Delta L}) / (2G), \quad (3)$$

In the PIA the ASE noise originates from both the amplification process and vacuum fluctuations entering the signal during transmission, while in the

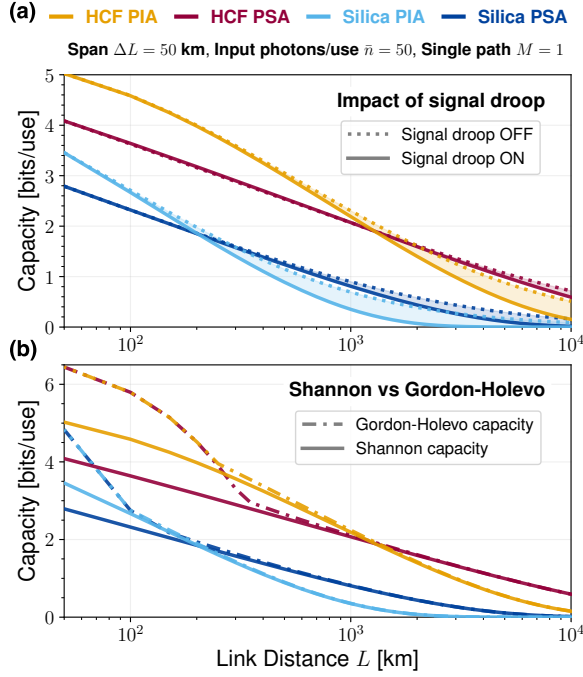


Fig. 2: Impact of signal droop (a) and quantum enhancement (b) on channel capacity simulated for four combinations of fiber and amplification type on an exemplary link.

PSA only the input noise from the previous span is amplified, hence the noise figure is improved at the cost of losing one quadrature.

The attainable rate of information transmission in a single use of a communication channel is conventionally given by the Shannon capacity. Importantly, the information rates can be in principle enhanced through the use of quantum receivers [6], utilizing e.g. collective detection [11]. The ultimate capacity limit follows from the Holevo bound [12] and for a standard additive Gaussian noise model takes a closed analytical form first introduced by Gordon [13]. When the excess noise is large, the Gordon-Holevo expression converges to the conventional Shannon capacity formula [14]. Since both PIA and PSA amplify the vacuum noise entering the signal during propagation, for large link lengths the Gordon-Holevo bound is expected to coincide with Shannon capacity.

For each scenario defined by parameters $(L, M, \alpha, \bar{n}_{\text{tot}}, \Delta L)$ we have optimized the Shannon capacity over gains G of amplifiers placed after each individual span. For PSA, we additionally optimized over the gain of a preamplifier at the channel input. We have enforced a total optical power constraint, meaning that after each amplifier the total energy including both signal and noise cannot exceed the initial value at the input of the link. Because of the noise accumulation during propagation, this constraint enforces signal droop and decrease of SNR [7].

It was shown in previous works [10, 15] that for a standard single-mode silica fiber PSA offers improved capacity for very large distances in the

continuous amplification regime. In practical scenarios, however, one is interested in optimizing the link with a discrete number of amplifiers as well as optimally dividing the input power between the multiple paths that constitute the link. In particular the latter may have a nontrivial effect, since on one hand multiple paths conventionally allow to transmit more information, but on the other hand a lower per-path-power may decrease the capacity due to combined effects of noise amplification and signal droop. In view of recent progress in HCF technology which promises much decreased attenuation factors, it is also important to analyze PSA and PIA enhanced bounds on information transmission rates under such favorable conditions.

Results

Fig. 2(a) plots Shannon capacity as a function of distance for a link with $\bar{n}_{\text{tot}} = 50$ input photons and $\Delta L = 50$ km span. We consider four cases: PIA or PSA amplification and either HCF ($\alpha = 0.04$ dB/km) or silica-based ($\alpha = 0.14$ dB/km) fiber type. For all cases the signal droop decreases capacity and the degradation is more important for PIA. This is in line with previous findings in the literature [7]; therefore, in the following signal droop is always included. For the same four cases it is seen in Fig. 2(b) that the Gordon-Holevo bound significantly exceeds Shannon capacity at short distances but they rapidly converge after a few hundreds of kilometers. As our work focuses mainly on long-haul systems, in the following we only report the Shannon capacity. It is notable, however, that for distances shorter than 500 km, the fundamental quantum limit on spectral efficiency is higher than the conventional Shannon capacity.

Fig. 3(a) illustrates the Shannon capacity vs average photon number per use $\bar{n}_{\text{tot}}$ for a single optical path, $M = 1$, corresponding to the four cases of: PSA (solid lines) and PIA (dashed lines) and either HCF or silica-based fiber types. Various span lengths ΔL are investigated while the total link distance is fixed to $L = 5000$ km. For easier reference, the upper x -axes translate $\bar{n}_{\text{tot}}$ into optical power of a transmitter operating in the optical C-band at a reference symbol rate of 300 GBaud. It is seen that in the low-power regime, PSA outperforms PIA, indicating that there is a capacity gain in using PSA to regenerate signals along individual optical paths, for instance when the total power is constrained and massive parallelism is employed. It is also noticeable that the cross-over point between the PIA and PSA capacity curves depends on the span length; moreover, the minimum power at which PIA outperforms PSA is lower for HCF.

Fig. 3(b,c) illustrates the total capacity over massively parallel channels vs total optical power, which is evenly shared among all paths. Fig. 3(b)

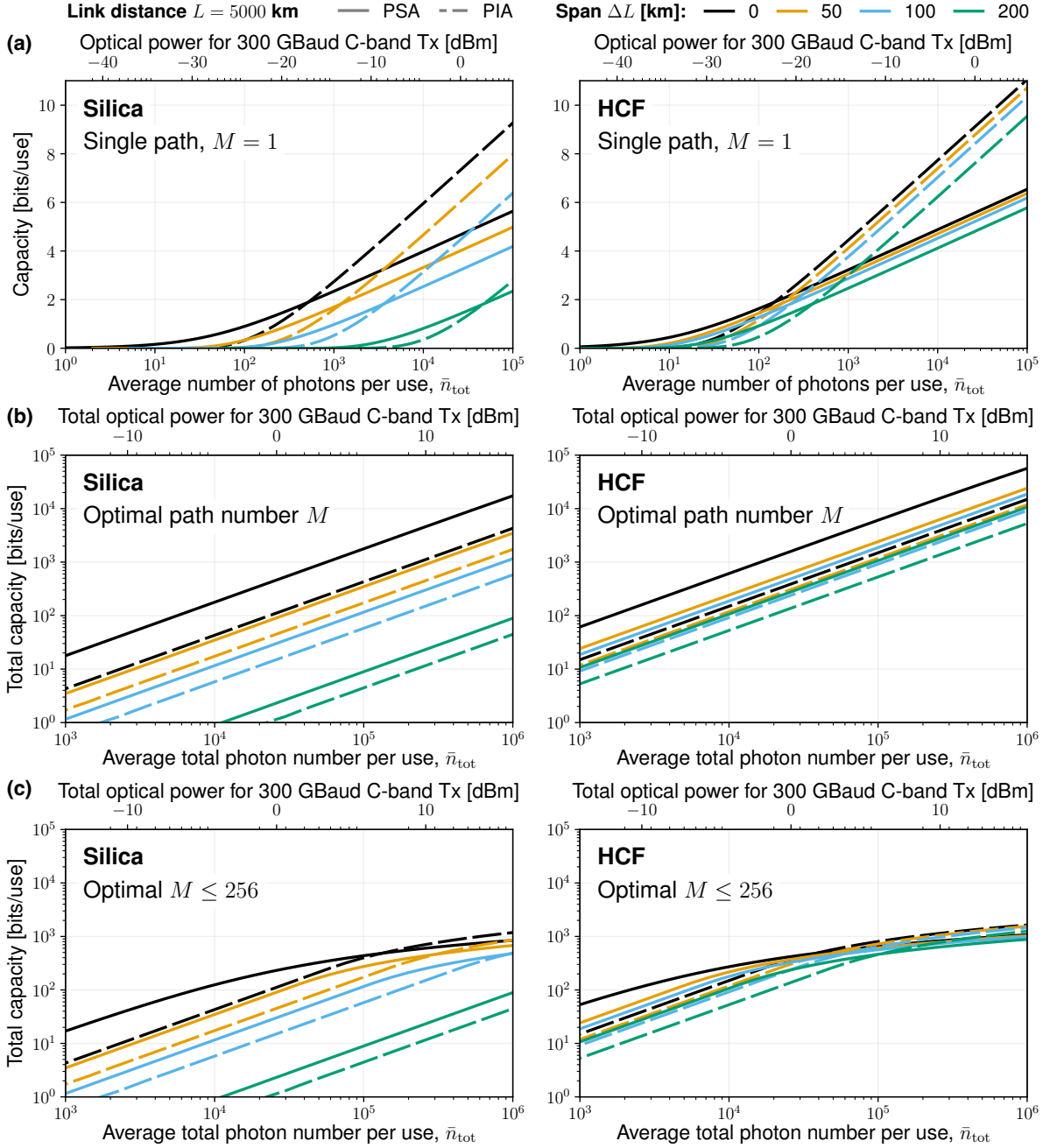


Fig. 3: Shannon channel capacity as a function of input photon number simulated for an exemplary 5000 km silica-based/HCF link. Different span lengths ΔL (black lines: distributed amplification, orange: 50 km span, blue: 100 km, green: 200 km) are pictured for both PIA (dashed lines) and PSA (solid lines). The upper axis additionally labels the input optical power in dBm units assuming 300 GBaud transmission in the optical C-band. (a) Single path case. (b) Impact of optimization over the number of parallel paths M . (c) Optimization with $M \leq 256$ constraint.

shows the results when the number of paths is optimized, whereas Fig. 3(c) illustrates the more practical case where the total number of parallel paths is constrained to be less than or equal to 256. As anticipated by the findings in Fig. 3(a), massive parallelism depicted in (b) results in PSA outperforming PIA in all cases, while in the path-constrained scenario shown in (c) there are cases where the PIA eventually outperforms PSA.

Conclusions

We provided a theoretical analysis of the fundamental achievable capacities in the future multi-

span and massively parallel long-haul fiber-optic transmission systems, investigating the benefits of phase-sensitive vs phase-insensitive amplifiers, as well as the impact of span length, signal droop, the choice of fiber type, HCF vs/conventional silica-based fiber, as well as the relevance of the Gordon-Holevo bound for capacity computations. We demonstrated that all these factors are inter-related and should be jointly optimized for future systems, and commented on the various regimes of operations where the use of PSA instead of PIA is recommended.

Acknowledgements

We acknowledge insightful discussions with P. Andrekson, M. Karlsson, and J. Schröder. This work was supported by the Foundation for Polish Science under the "Quantum Optical Technologies" project carried out within the International Research Agendas programme co-financed by the European Union under the European Regional Development Fund and by the European Union's Horizon Europe research and innovation program under the project Quantum Secure Networks Partnership (QSNP, grant agreement No 101114043). K.Ł. acknowledges support by the Polish Ministry of Education and Science under the project "Quantum strategies in communication through noisy optical channels", contract no. PN/01/0204/2022 carried out within the "Pearls of Science" program.

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